

CHATLA VARUN

Software Engineer (MLOps /DevOps)

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Summary

Certified Kubernetes Administrator (CKA) with **4.5 years of experience** in Kubernetes administration, **DevOps and MLOps engineering**, infrastructure management, and delivery operations. Proficient in leveraging **Docker, Kubernetes, Jenkins, and GitHub Actions** to streamline CI/CD pipelines and ML workflows, reducing deployment times by up to **40%** and improving infrastructure scalability by 30%. Experienced in building and operating **end-to-end MLOps pipelines** including model training, experiment tracking, versioning, deployment, and scalable model serving using **MLflow, DVC, Kubeflow, KServe, and AWS SageMaker**, ensuring reproducibility, reliability, and production-ready ML systems.

Technical Skills

- **CI/CD & DevOps:** Jenkins, GitHub Actions, Terraform, Ansible
- **Containers & Orchestration:** Docker, Kubernetes
- **Cloud:** AWS, Azure
- **MLOps Tools:** DVC, MLflow, Kubeflow, KServe, SageMaker Studio
- **Programming:** Python (primary), Java (secondary)
- **Databases & APIs:** PostgreSQL, FastAPI, Flask
- **Version Control:** GitHub, Bitbucket
- **Observability:** Prometheus, Grafana, CloudWatch
- **Security:** IAM, Secrets Manager, Vault, Trivy
- **Others:** Jira, Confluence

Professional Certification and Training

- Certified Kubernetes Administrator Certificate ID-**LF-8lxwy4kpmq**

Professional Experience

Cyient Ltd — June 2021 – Present

Software Engineer (MLOps /DevOps) — Client: TomTom

MLOps Responsibilities

- Designed and implemented an end-to-end MLOps workflow, covering model training, versioning, deployment, serving, scaling, and lifecycle management.
- Built and productionized ML models with a clear separation of concerns between model logic, serving layer, and infrastructure, following MLOps best practices.
- Implemented custom model serving using Flask + Gunicorn (WSGI) to understand low-level model serving before adopting managed platforms.
- Containerized ML workloads using Docker, ensuring reproducible builds and immutable model artifacts across environments.
- Deployed ML models on Kubernetes clusters, implementing Deployments, Services, and Ingress for scalable and resilient inference.
- Implemented KServe-based model serving, leveraging standardized inference APIs and cloud-native model deployment patterns.
- Worked hands-on with KServe predictors, understanding model formats, storage backends, autoscaling behavior, and request/response contracts.
- Compared and evaluated custom Flask-based serving vs KServe-managed serving, identifying trade-offs in scalability, operability, and maintenance.
- Integrated external model artifact storage (S3 and HTTP-based endpoints) for decoupled model deployment in Kubernetes.
- Implemented AWS SageMaker end-to-end workflows, covering model training, deployment, and managed inference endpoints.
- Understood and applied SageMaker Domain, User Profiles, and IAM execution roles, aligning ML access control with enterprise security practices.

- Designed IAM-based role separation for Data Scientists, ML Engineers, and MLOps Engineers within SageMaker environments.
- Deployed and tested SageMaker-hosted inference endpoints, validating real-time prediction workflows.
- Built auto-scaling model-serving architectures using AWS Auto Scaling Groups and Application Load Balancers.
- Designed high-availability inference architectures using load balancers to distribute prediction traffic across replicas.
- Implemented cloud-native bootstrapping using EC2 user-data, enabling automated ML service provisioning without manual intervention.
- Applied MLOps automation principles using CI concepts to automate model training, artifact generation, and deployment readiness.
- Followed production ML design principles, including stateless inference, horizontal scalability, and environment reproducibility.
- Gained hands-on experience with MLOps tooling across VMs, Kubernetes, and managed cloud services, understanding when to use each approach.
- Developed a strong understanding of real-world MLOps responsibilities, including model reliability, deployment safety, infrastructure automation, and operational scalability.

DevOps Responsibilities

- Installed and configured Docker, including Docker image creation and container orchestration.
- Managed Kubernetes resources such as Pods, Deployments, and Rolling Updates for seamless application scaling.
- Deployed and maintained microservices on AWS ECS and EKS, ensuring high availability and performance.
- Designed and developed CI/CD pipelines using Jenkins and GitHub Actions for efficient software delivery.
- Upgraded Kubernetes clusters with zero-downtime strategies, optimizing performance and reliability.
- Proficient in leveraging AWS services, including EC2, S3, RDS, EBS, VPC, IAM, ELB, CloudWatch, CloudFormation, Route 53, and Security Groups.
- Resolved critical issues related to Kubernetes control plane failures and container images, minimizing downtime.
- Authored Docker files and advised clients on advanced Docker best practices enhancing containerized application performance.
- Developed Proof of Concepts (POC) for infrastructure automation using Terraform and secured sensitive data with Vault.
- Managed RCA and problem management workflows, achieving a 30% reduction in production issues through proactive measures.
- Configured commit-based Jenkins jobs using webhooks to streamline the CI/CD workflow.
- Configured remote proxy repositories for Conan package management to streamline dependency management and version control.
- Streamlined repetitive tasks using shell scripting and crafted Ansible playbooks for tool installations and artifact deployments.
- Provisioned infrastructure using **Terraform**, creating reusable and scalable modules for cloud resources like EC2, RDS, and S3.
- Integrated **Terraform** into CI/CD pipelines to automate environment setup and teardown, improving deployment efficiency.
- Automated server configuration and application deployments using Ansible, ensuring consistency across environments.
- Engineered containerized solutions using Docker, including multi-stage Docker files and private registry configurations.
- Facilitated AWS activities such as creating EC2 instances, configuring security group rules, creating S3 buckets, and managing snapshots.
- Designed and delivered CI/CD pipelines with Jenkins, integrating tools like SonarQube, Docker, Ansible, and Kubernetes.
- Created Helm charts for deploying applications on Kubernetes and managing resources effectively.
- Handled Helm operations including installation, linting, upgrades, rollbacks, and customization of templates.
- Monitored and troubleshot Jenkins jobs, performed root cause analysis, and addressed Maven build warnings and errors.
- Transitioned Jenkins freestyle jobs to Pipeline jobs, enhancing automation and build efficiency.
- Optimized existing processes and documented workflows for easy knowledge transfer.
- Innovated Proof of Concepts (POCs) for infrastructure automation using Terraform and secured sensitive data with Vault.
- Configured Jenkins build jobs for multiple environments, ensuring continuous integration.